

DRAEM - ICCV 2021

A Discriminatively Trained Reconstruction Embedding for Surface Anomaly Detection – Vitjan Zavrtanik, Matej Kristan, Danijel Skočaj

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- Detection and localization of anomalies in the image space

Introduction

Surface Anomaly Detection

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- Therefore, methods that only rely on normal data are preferred

Problem Statement

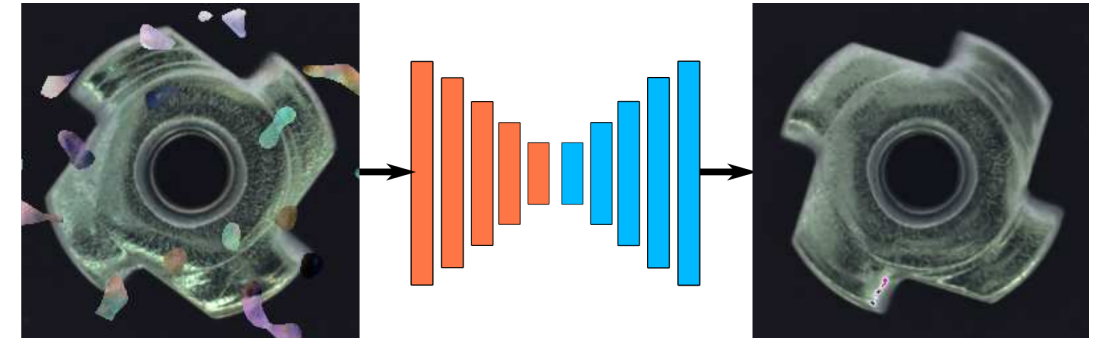
Current Situation and Related Work

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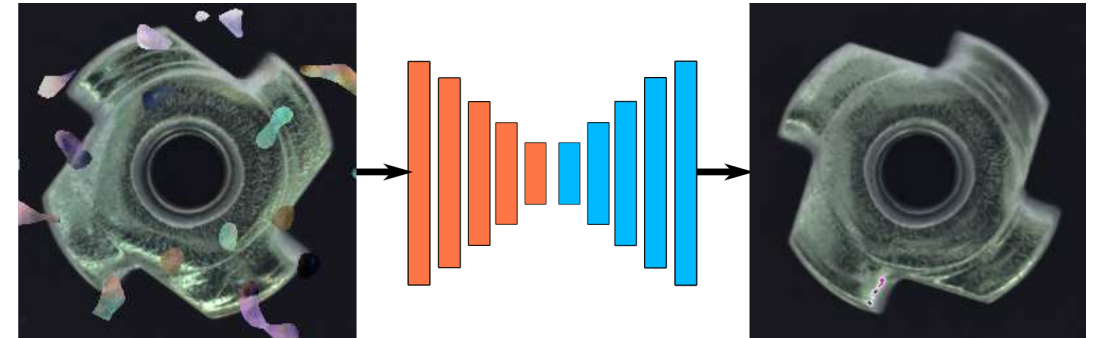
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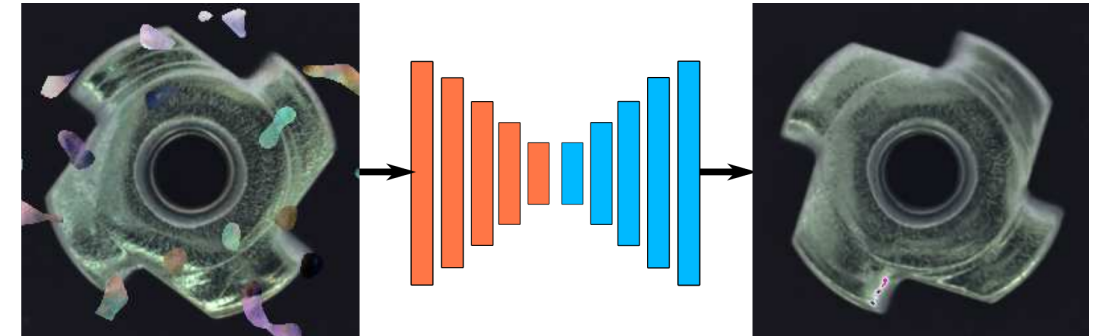
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 - Often suffer from unsharp segmentation
 - Overfit to synthetic anomalies

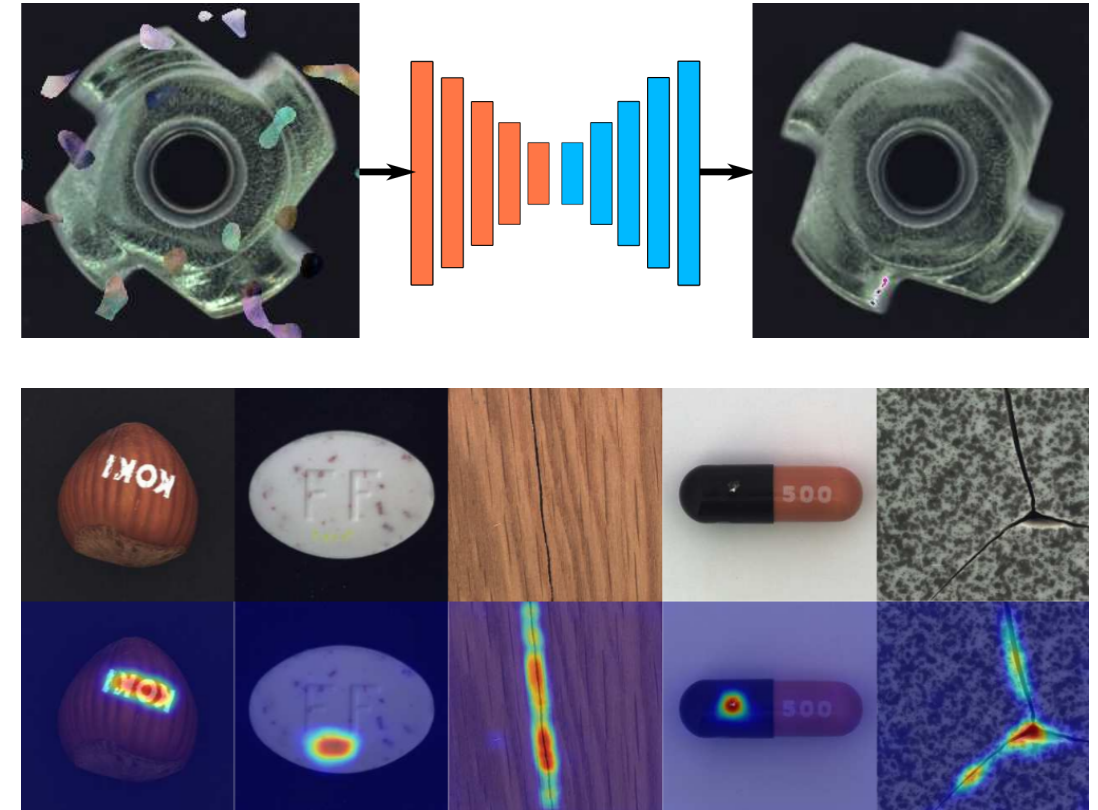
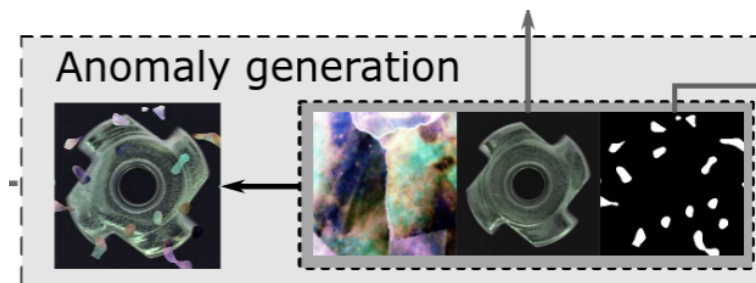


Figure: Unsharp segmentation maps from discriminative methods.

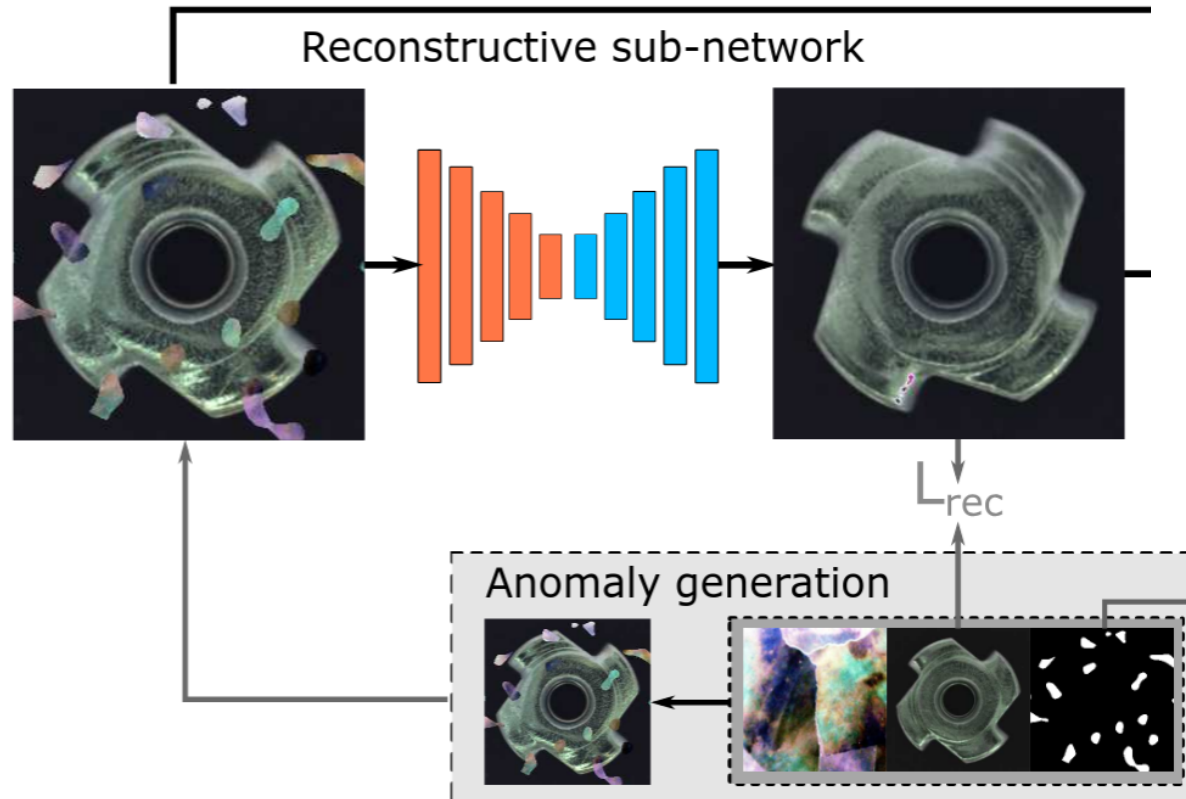
Framework Overview

How it Works



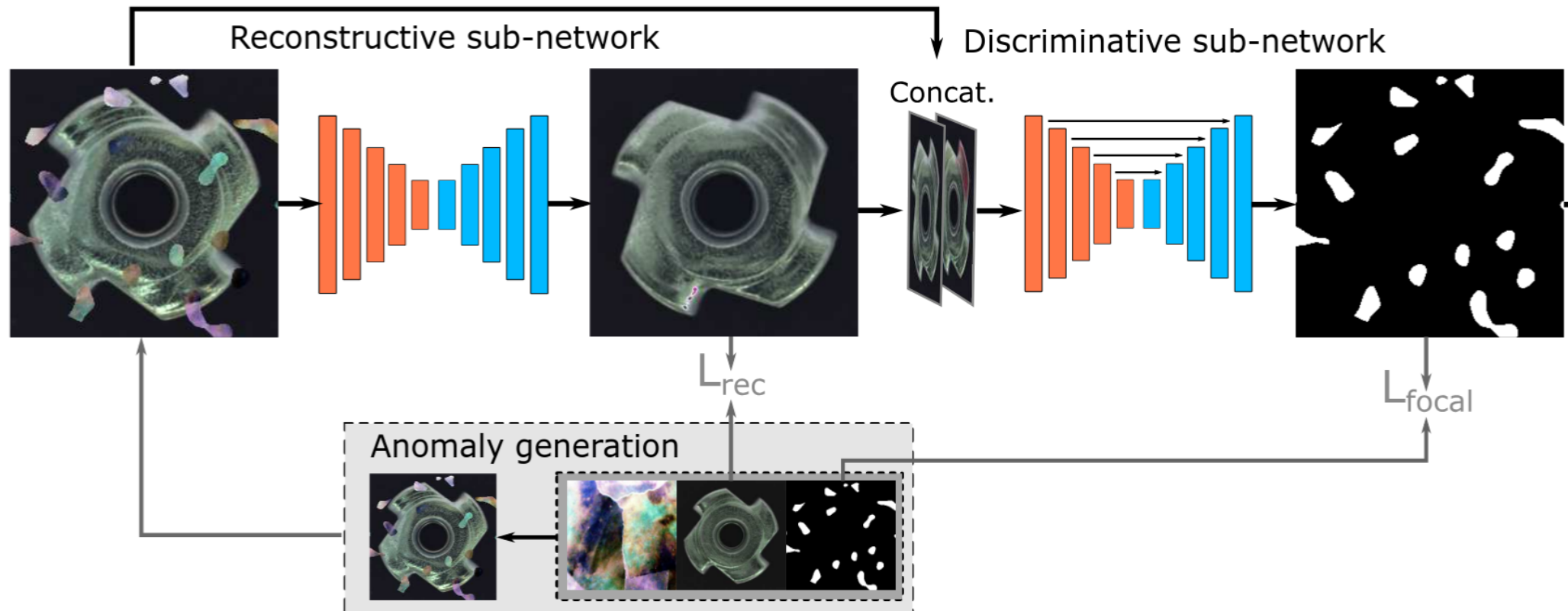
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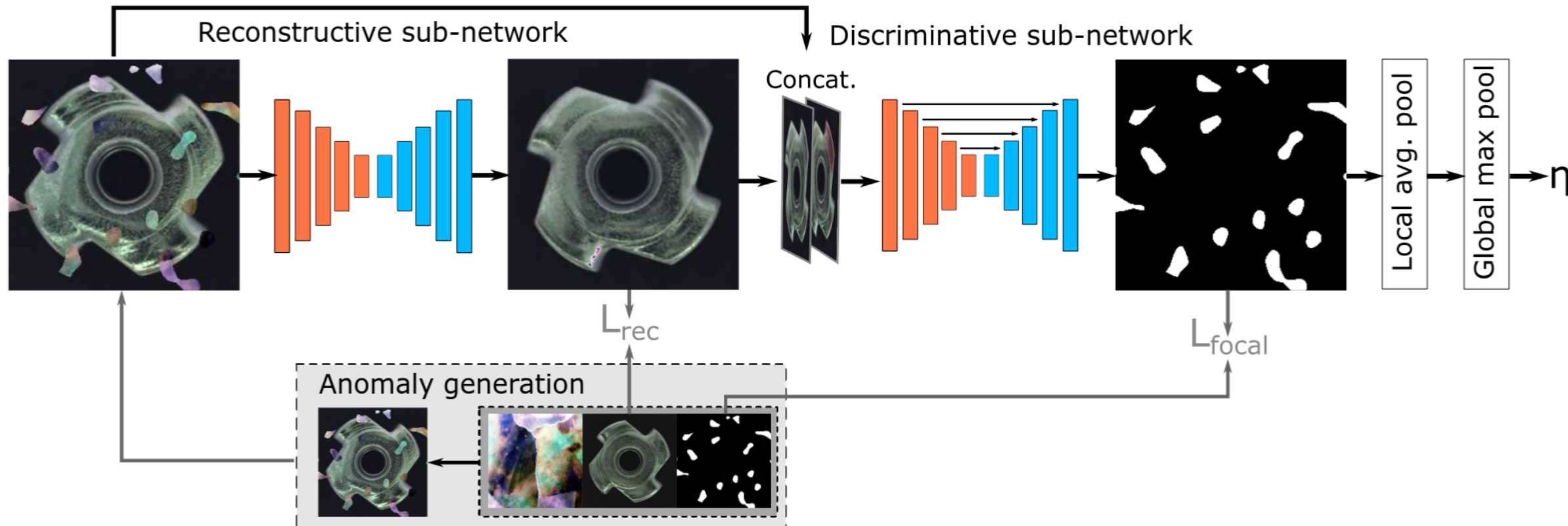
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Performance Comparison

DRÆM vs State-of-the-Art

Method	Image AUROC	Pixel AUROC	Pixel AP
US	87.7%	93.9%	45.5%
RIAD	91.7%	94.2%	48.2%
PaDim	<u>95.5%</u>	97.4%	<u>55.5%</u>
DRÆM	98.0%	<u>97.3%</u>	68.4%

Table: Results on MVTec Dataset

Performance on DAGM Dataset

Unsupervised vs. Supervised Methods

Method	AUROC	TPR	TNR	CA
Unsupervised				
US	72.5	72.6	65.3	66.2
RIAD	78.6	79.2	69.1	70.4
PaDim	95.0	83.3	97.5	95.7
DRÆM	99.0	96.5	99.4	98.5

Table: DRÆM on DAGM dataset: outperforming unsupervised methods and on par with fully supervised ones.

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Supervised				
CADN	—	—	—	89.1
Racki et al.	99.6	99.9	99.5	—
Lin et al.	99.0	99.4	99.9	—
Bozic et al.	100	100	100	100

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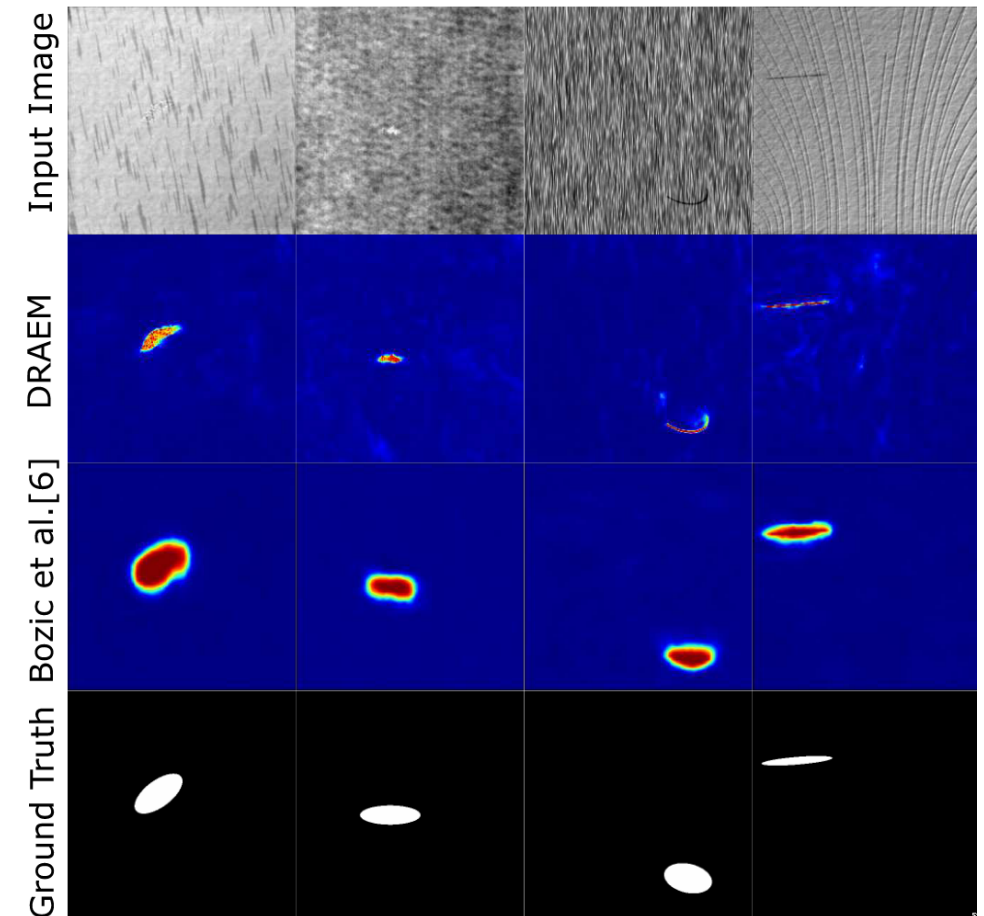


Figure: DRÆM vs supervised SOTA (Bozic et al.) on DAGM.

Key Findings & Application & Conclusion

The Value of DRÆM

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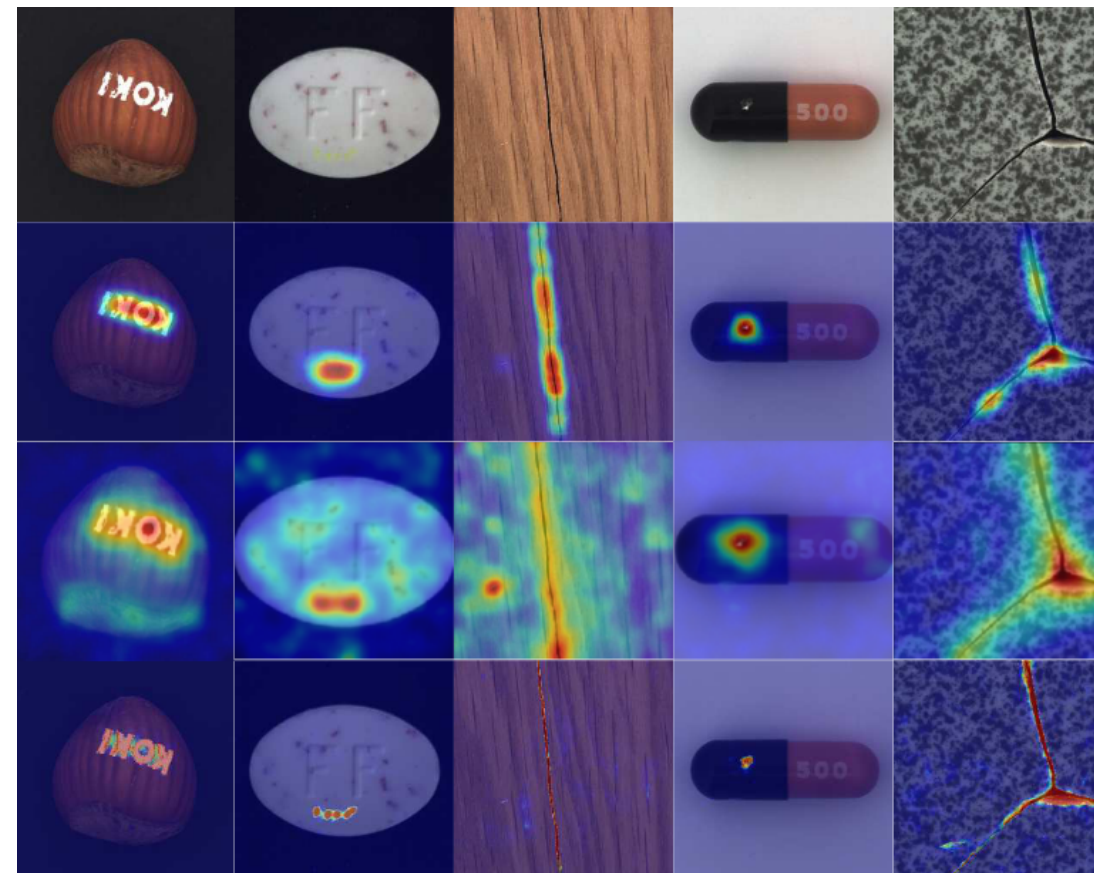


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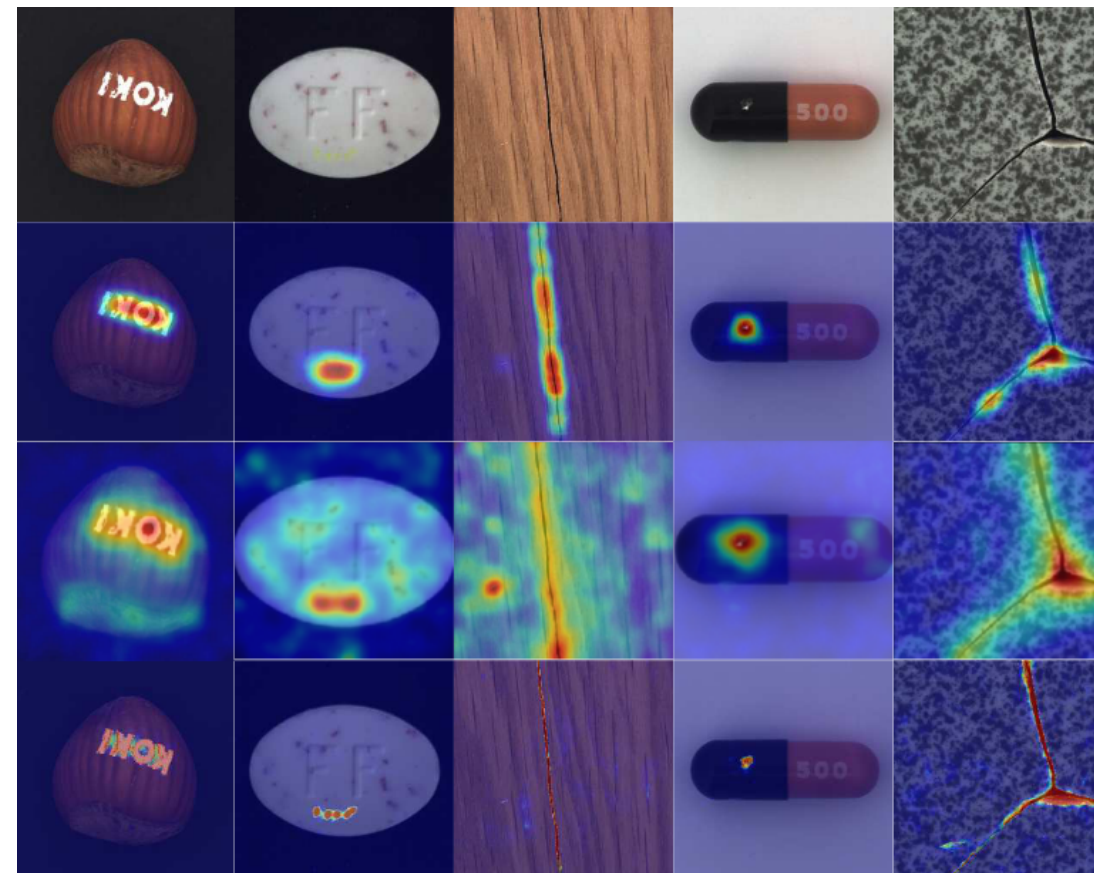
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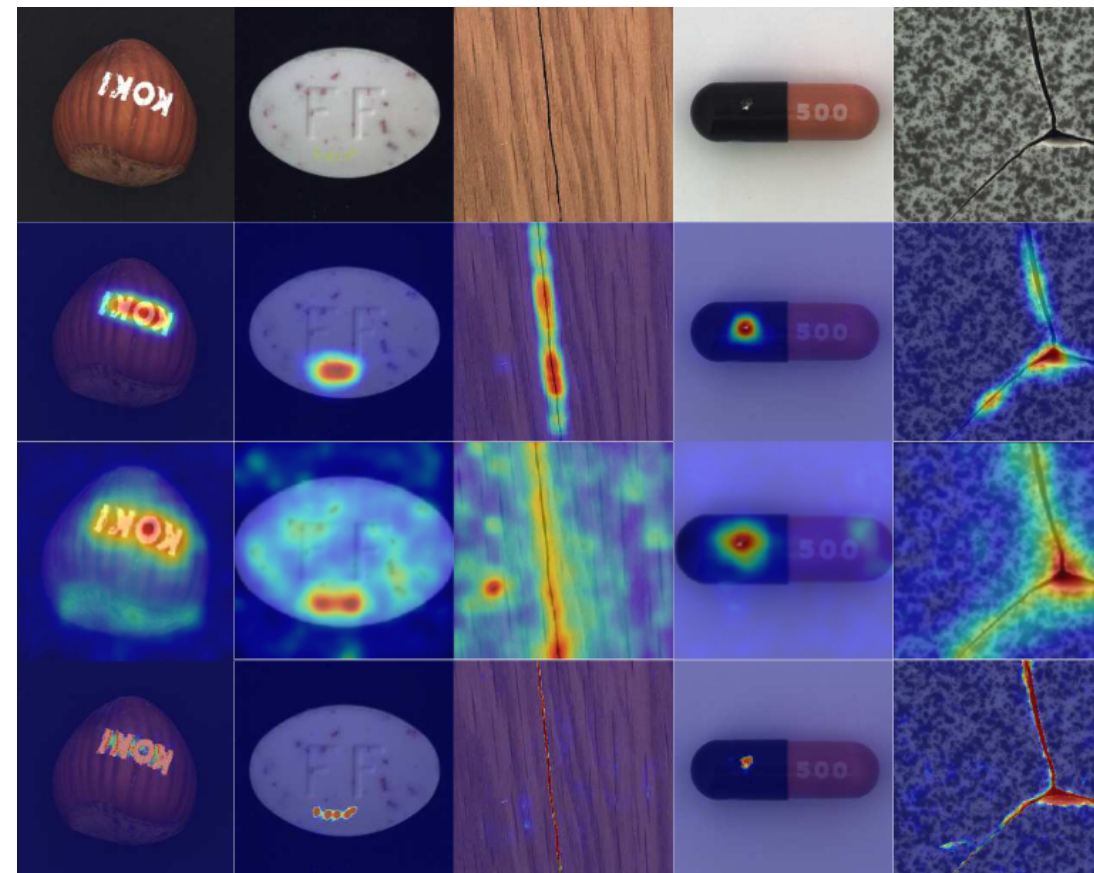
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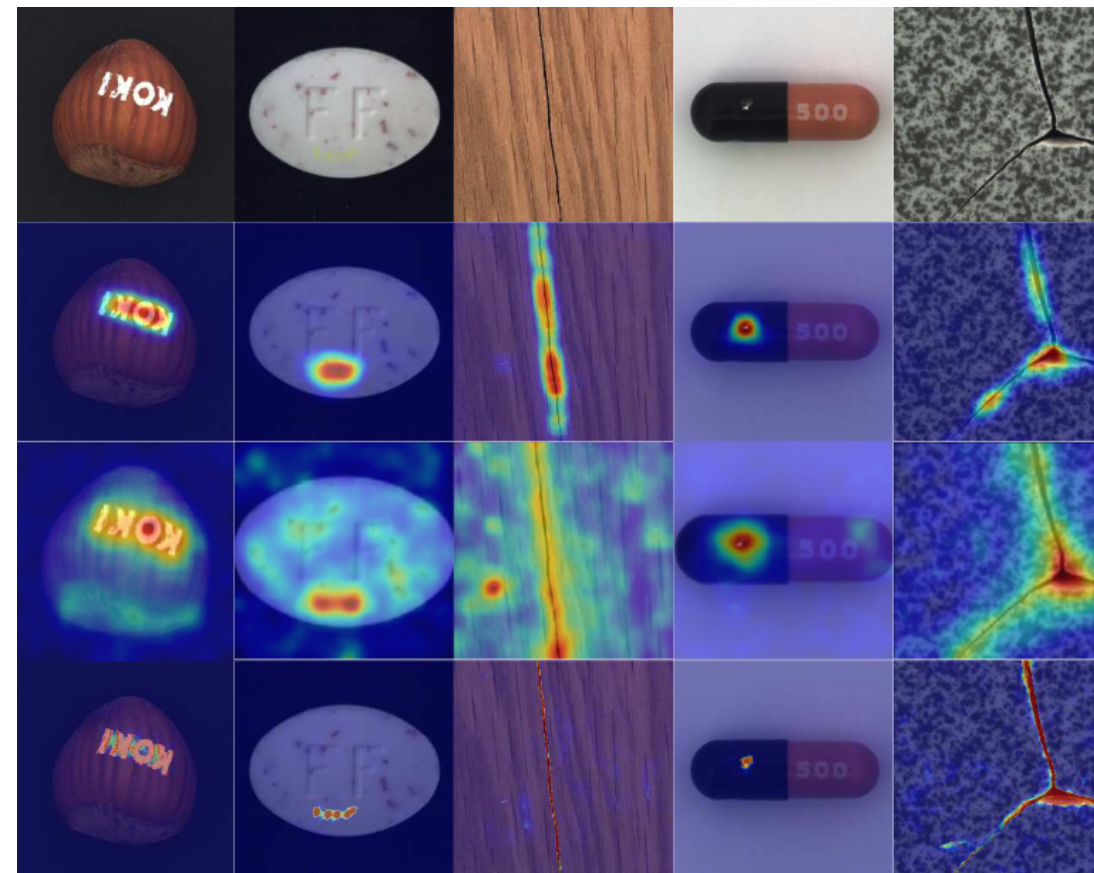
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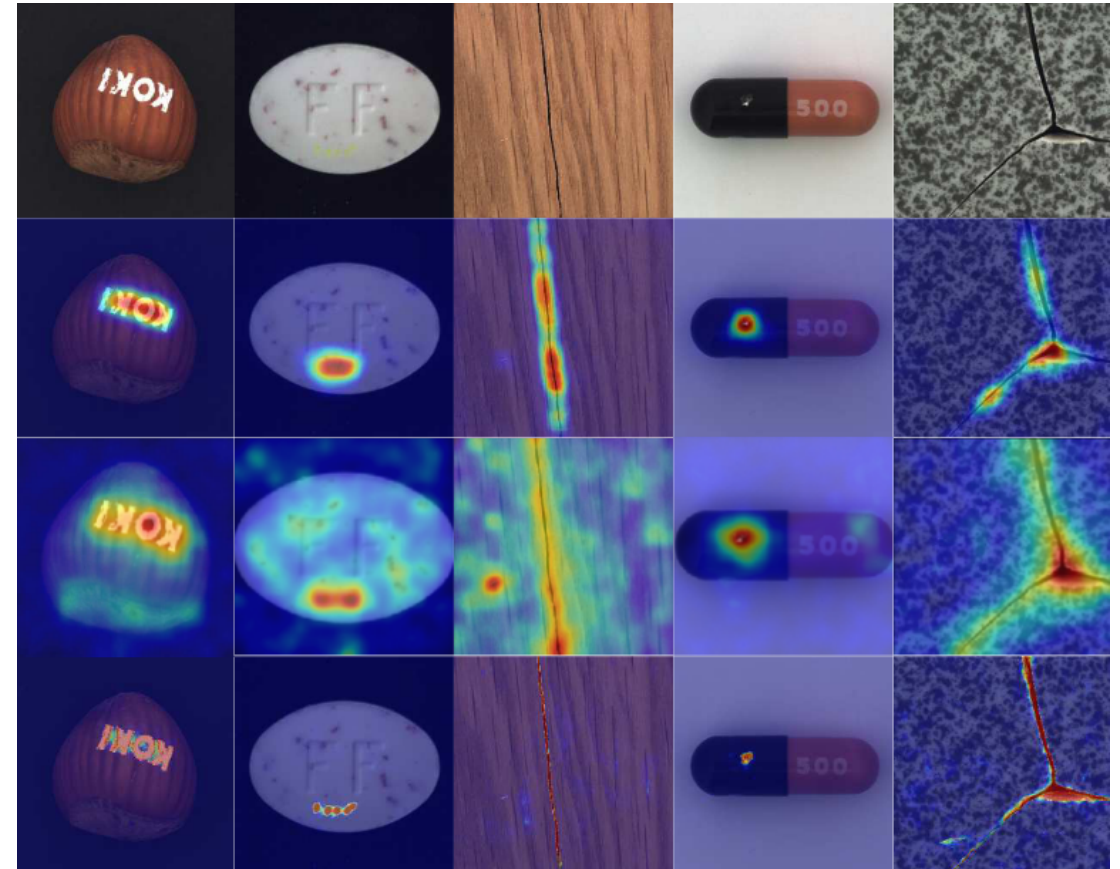
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- High quality segmentation can help downstream tasks
 - Estimate the size of impacted area
 - Distinguish between crack, corrosion or just paint
- We don't need expensive hand-labeled data
- We only need just-out-of-distribution patterns for anomaly generation



The Provided Code

Complete and Good?

× No code for experiments on DAGM dataset

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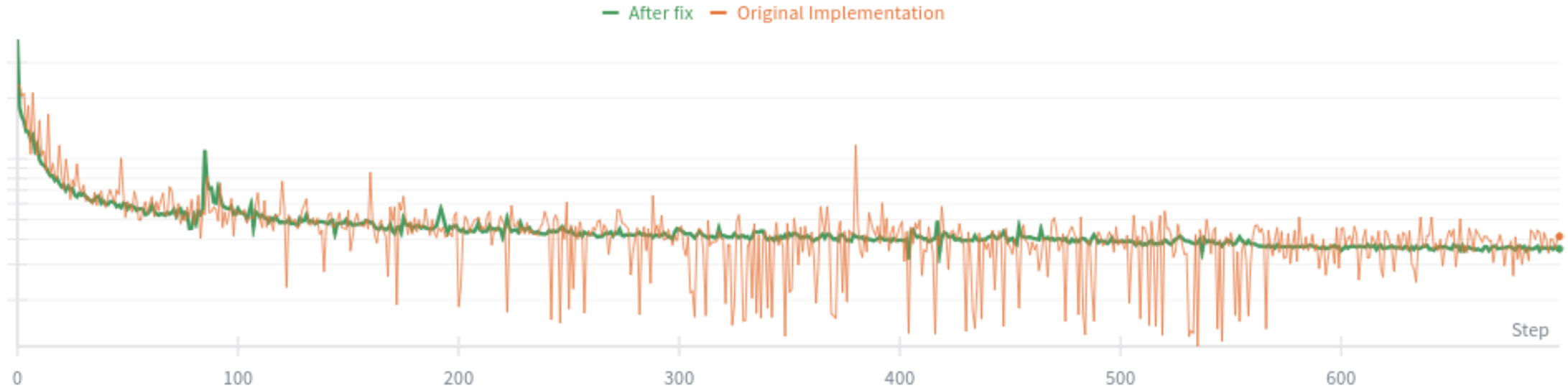


Figure: Loss smoothing by weighting last batch of each epoch

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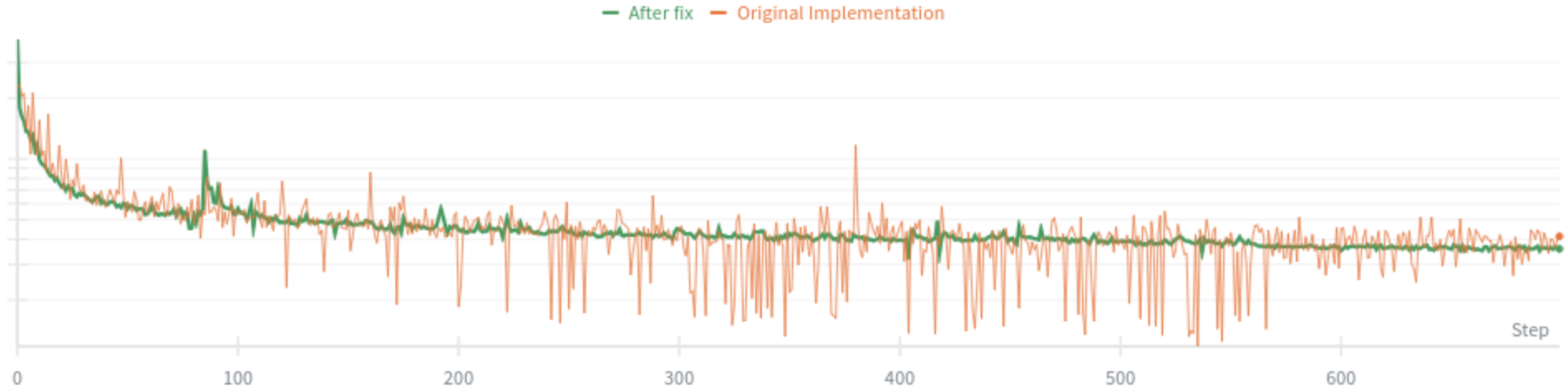


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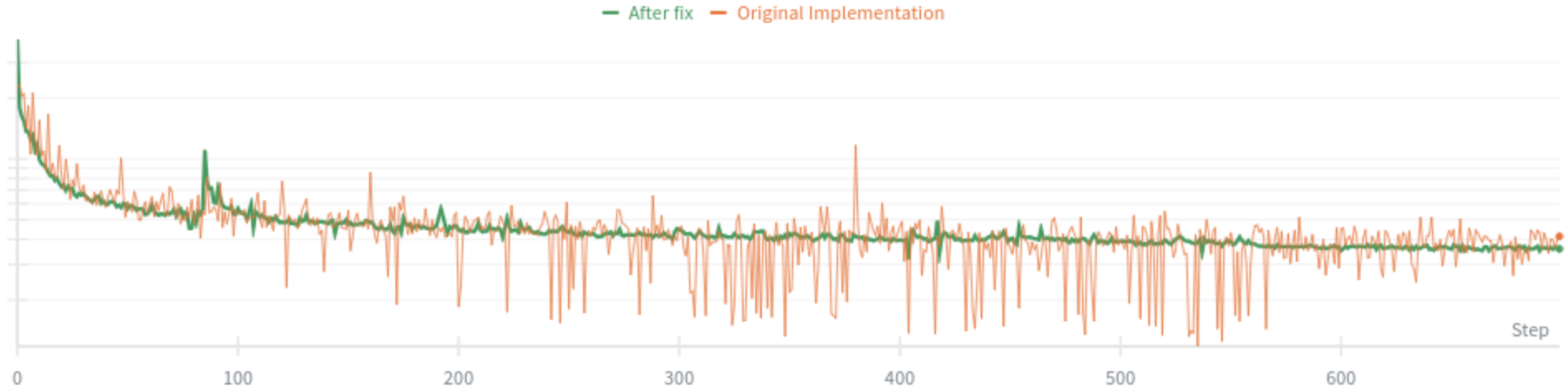


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- ✓ Well readable code

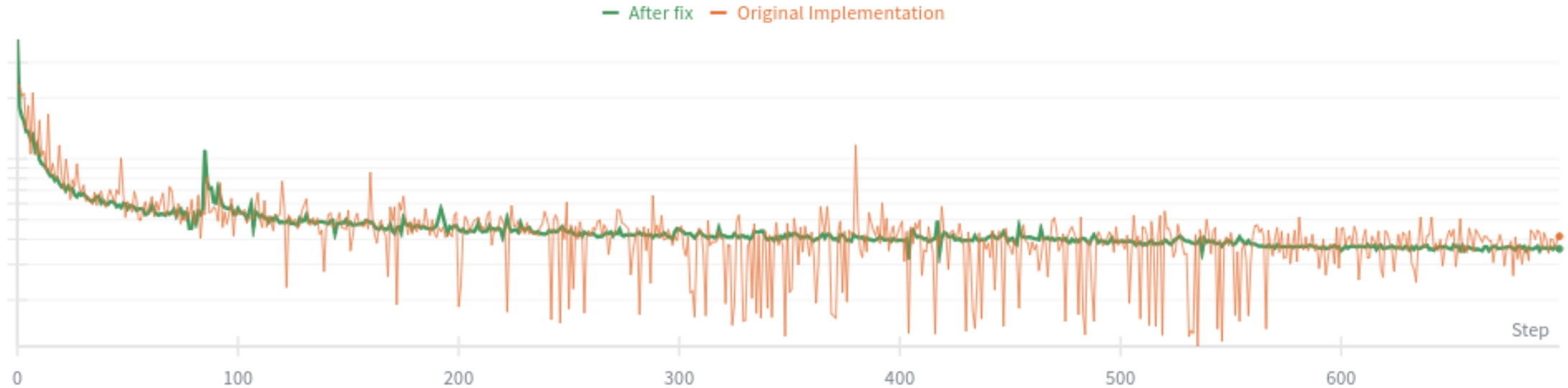


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- ✓ Download script for MVTec and DTD datasets

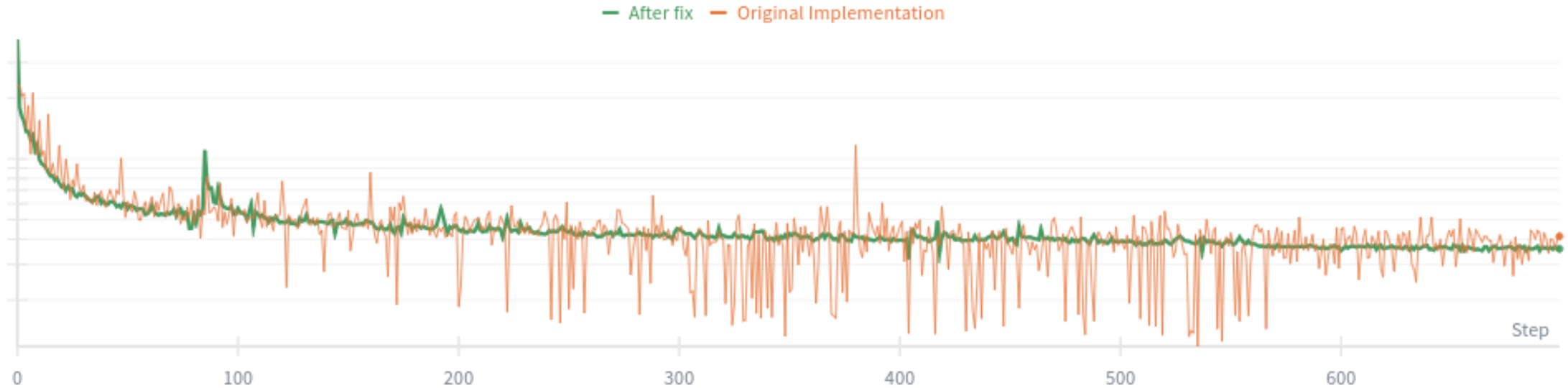


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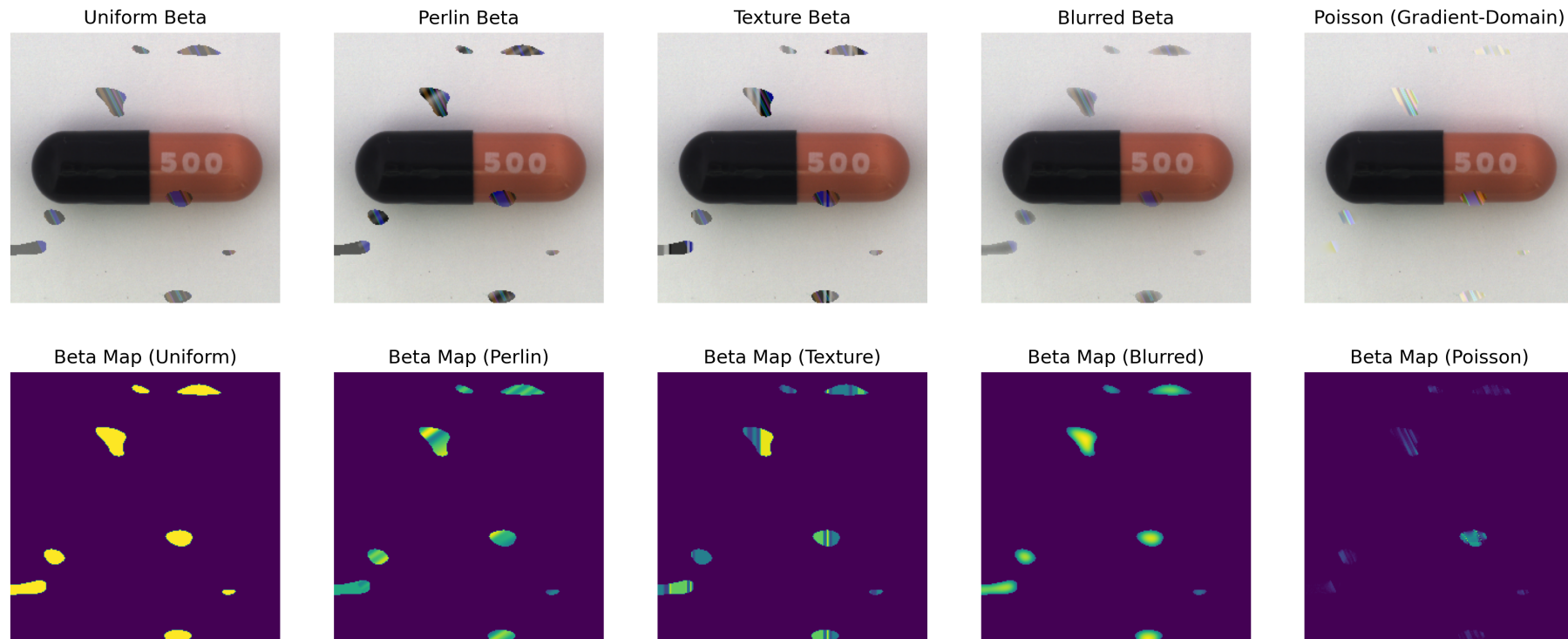


Figure: Different blending methods (from left to right): uniform beta, perlin beta, texture beta, blurred beta, poisson blending

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DRÆM-Texture	98.0%	97.2%	69.1%
DRÆM-Perlin	98.2%	97.3%	69.5%
DRÆM-Poisson	<u>99.1%</u>	<u>97.8%</u>	<u>71.6%</u>

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DRÆM-Perlin	98.2%	97.3%	69.5%
DRÆM-Poisson	<u>99.1%</u>	<u>97.8%</u>	<u>71.6%</u>
DRÆM-Poisson+Perlin	99.2%	98.0%	73.6%

Table: Results on MVTec Dataset

Thank you for your attention

The background of the slide is a solid blue color. In the lower half, there is a decorative graphic consisting of numerous thin, concentric, curved lines that create a sense of motion or a ripple effect, starting from the bottom and curving upwards towards the center.